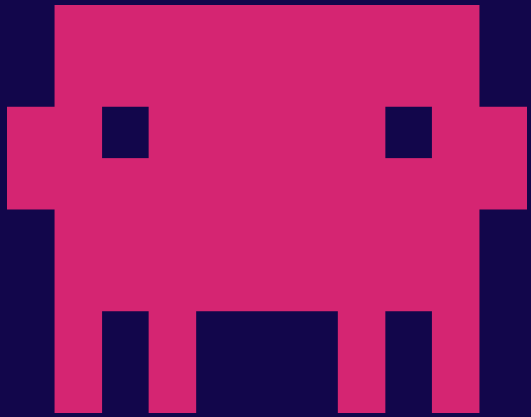


Cooking with Claude

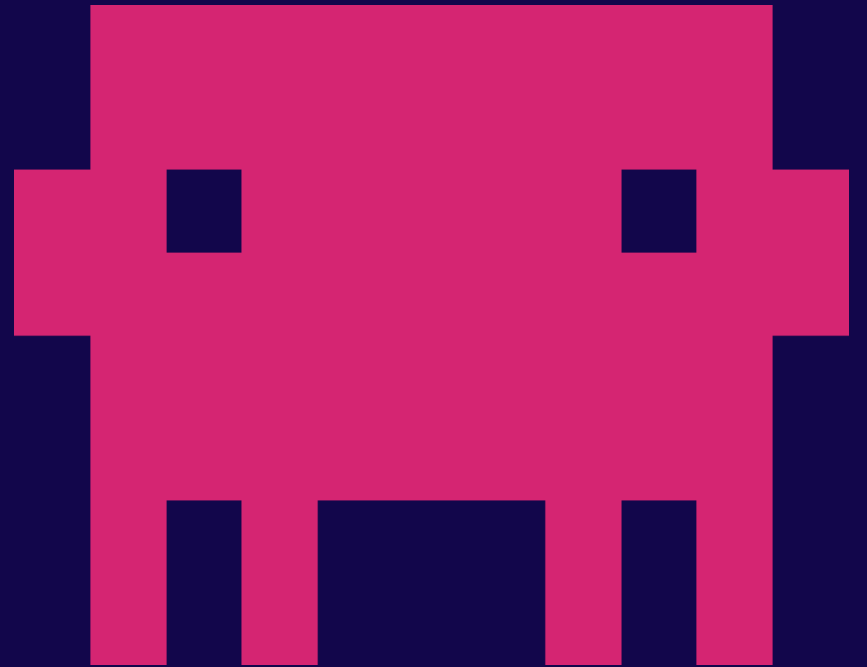


By Lumos

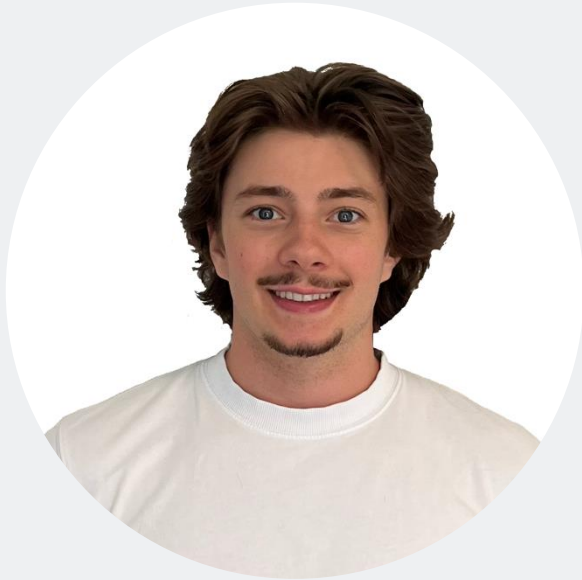


Agenda for Today

1. Introduction
2. Claude Basics
3. Context Window
4. Claude Features
5. Skills, MCP & Agents
6. Claude Code in Action



Presented by



Michael Bösch

Born Vibe Coder



Dominik Kanjuh

Knows a bit about software

Lumos- We bring curious students from all universities together to:



- ▶ Build stuff around tech.
- ▶ Solve industry problems in collaboration with companies.
- ▶ Organize Hackathons, our workshops like this one.





Why are we sitting
Here today?

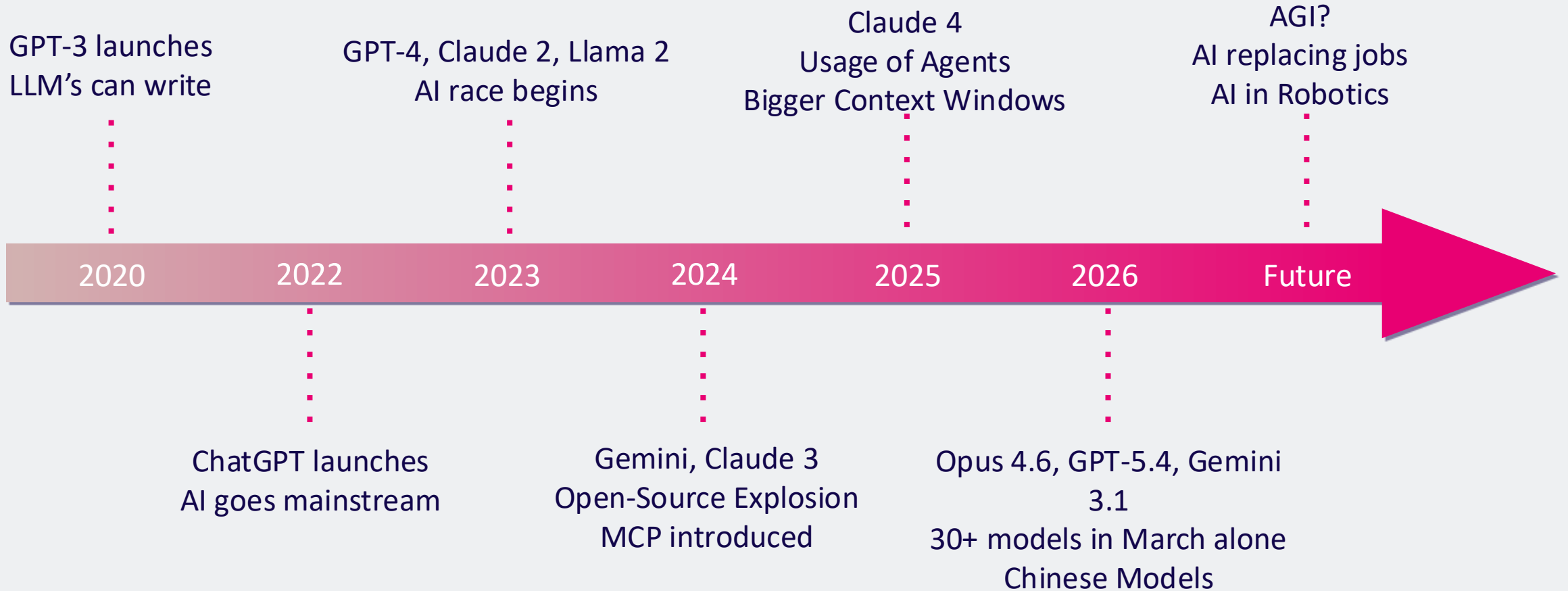


"This tsunami is coming at us. It's so close we can see it on the horizon."



Dario Amodei
CEO, Anthropic

Evolution of AI

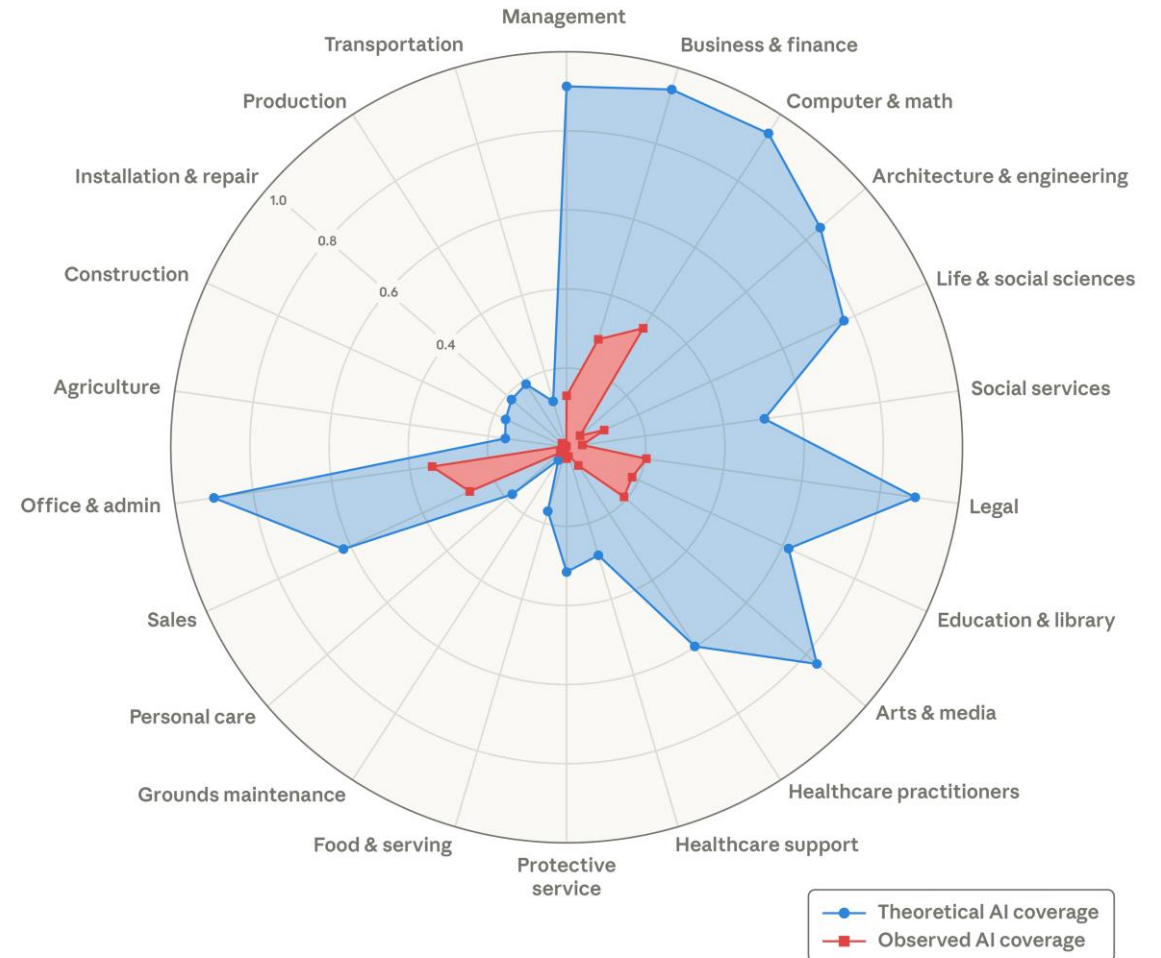


Anthropic Industry Report



- ▶ 14% decline in entry-level hiring in exposed roles
- ▶ Least exposed: trades (cooks, mechanics, electricians)
- ▶ AI skills compound: early adopters are pulling ahead

Theoretical capability and observed usage by occupational category

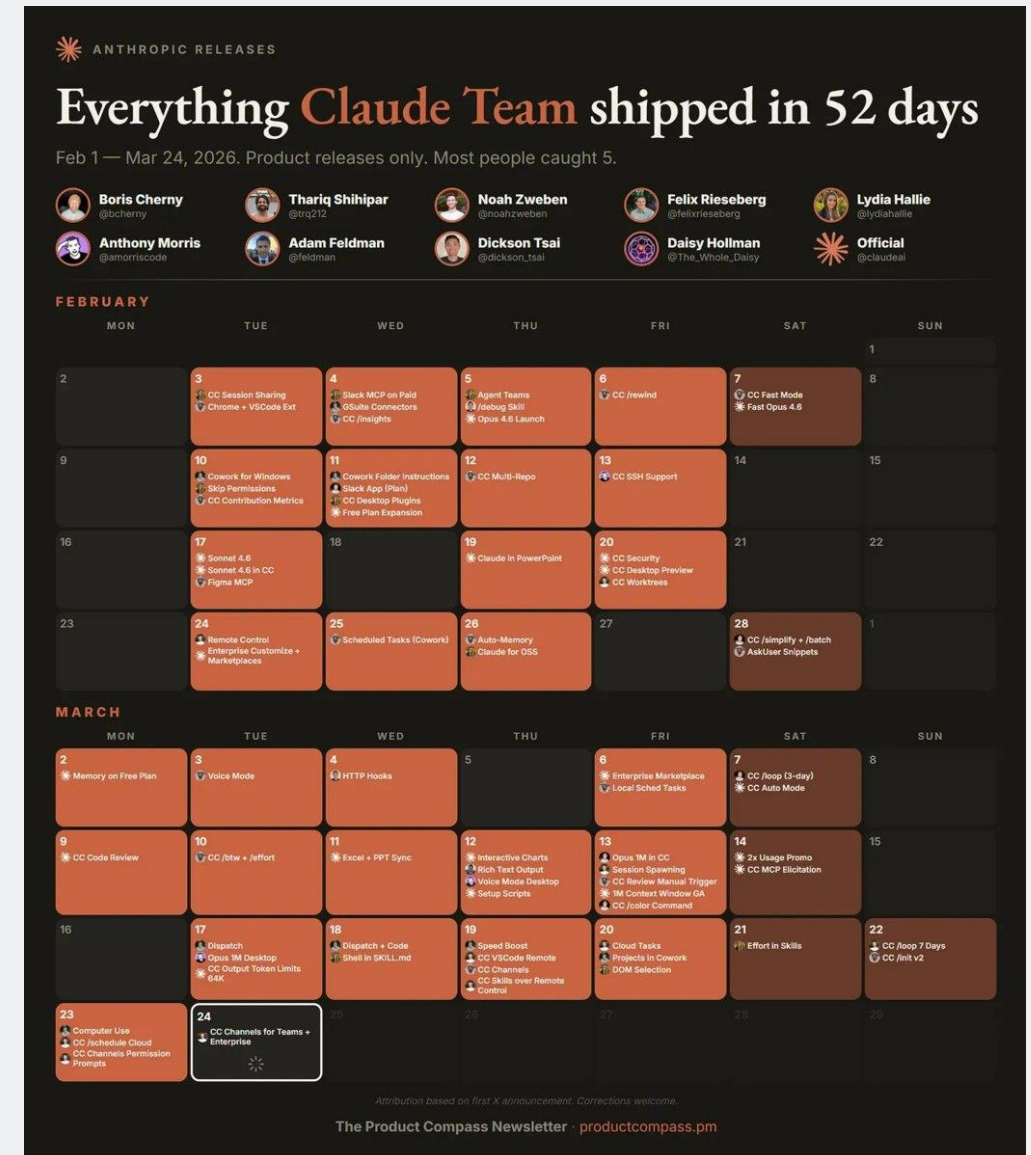


Space is moving so fast...



Just the last 4 weeks:

- ▶ Mar 17: GPT-5.4 launches
- ▶ Mar 20: Gemini 3.1 with 2M context
- ▶ Mar 24: Sora shutdown
- ▶ Apr 1: Q1 AI Funding hits \$300B
- ▶ Apr 2: Google releases Gemma 4
- ▶ Apr 7: Claude Mythos Preview





- "Cursor is the best for vibe coding"
- "Claude code is better. Have your projects made any money?"
- "No, have yours?"
- "No"





Claude Basics

What is Anthropic & Claude?



Anthropic

Founded 2021

Valuation: \$380B

IPO planned Oct 2026



Claude

Product Series

Frontier Models

“Good” Public Perception



The Model Selection

Opus 4.6 - The Head Chef

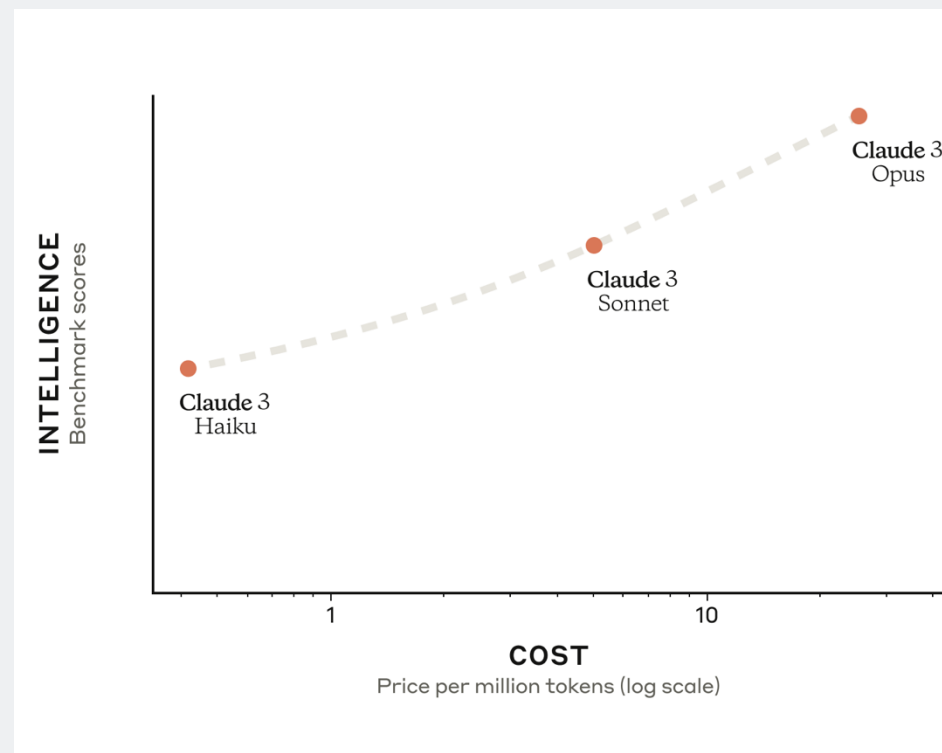
- ▶ Complex reasoning
- ▶ 1M token context, \$5/\$25 per 1M tokens

Sonnet 4.6 - The Sous Chef

- ▶ Daily driver
- ▶ 1M token context, \$3/\$15 per 1M tokens

Haiku 4.5 - The Prep Cook

- ▶ Fast, cheap
- ▶ 200K token context, \$1/\$5 per 1M tokens



LLM Arena Benchmark



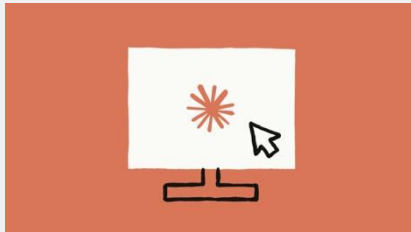
Overall

Rank ↕	Model ↕	Score ↓
1	 claude-opus-4-6-thinking ↗ Anthropic · Proprietary	1503 ±5
2	 claude-opus-4-6 Anthropic · Proprietary	1497 ±5
3	 gemini-3.1-pro-preview Google · Proprietary	1493 ±5
4	 grok-4.20-beta1 xAI · Proprietary	1490 ±7
5	 gemini-3-pro Google · Proprietary	1486 ±4
6	 gpt-5.4-high OpenAI · Proprietary	1484 ±7
7	 grok-4.20-beta-0309-reasoning xAI · Proprietary	1480 ±6
8	 gpt-5.2-chat-latest-20260210 OpenAI · Proprietary	1477 ±5
9	 grok-4.20-multi-agent-beta-0309 xAI · Proprietary	1475 ±6
10	 gemini-3-flash Google · Proprietary	1474 ±4

Coding

Rank ↕	Model ↕	Score ↓
1	 claude-opus-4-6-thinking Anthropic · Proprietary	1555 ±11
2	 claude-opus-4-6 Anthropic · Proprietary	1546 ±10
3	 gpt-5.4-high OpenAI · Proprietary	1532 ±14
4	 gemini-3.1-pro-preview Google · Proprietary	1531 ±9
5	 claude-opus-4-5-20251101-thinking-32k Anthropic · Proprietary	1531 ±7
6	 grok-4.20-beta1 ↗ xAI · Proprietary	1524 ±14
7	 grok-4.20-multi-agent-beta-0309 xAI · Proprietary	1523 ±12
8	 claude-sonnet-4-6 Anthropic · Proprietary	1522 ±12
9	 grok-4.20-beta-0309-reasoning xAI · Proprietary	1522 ±13
10	 glm-5.1 Z.ai · MIT	1520 ±19

The Claude Ecosystem



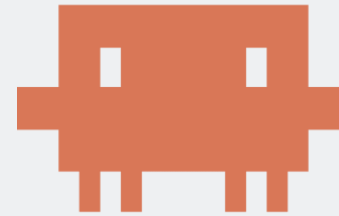
Claude Desktop

Regular Chat



Claude Cowork

Desktop agent



Claude Code

CLI coding



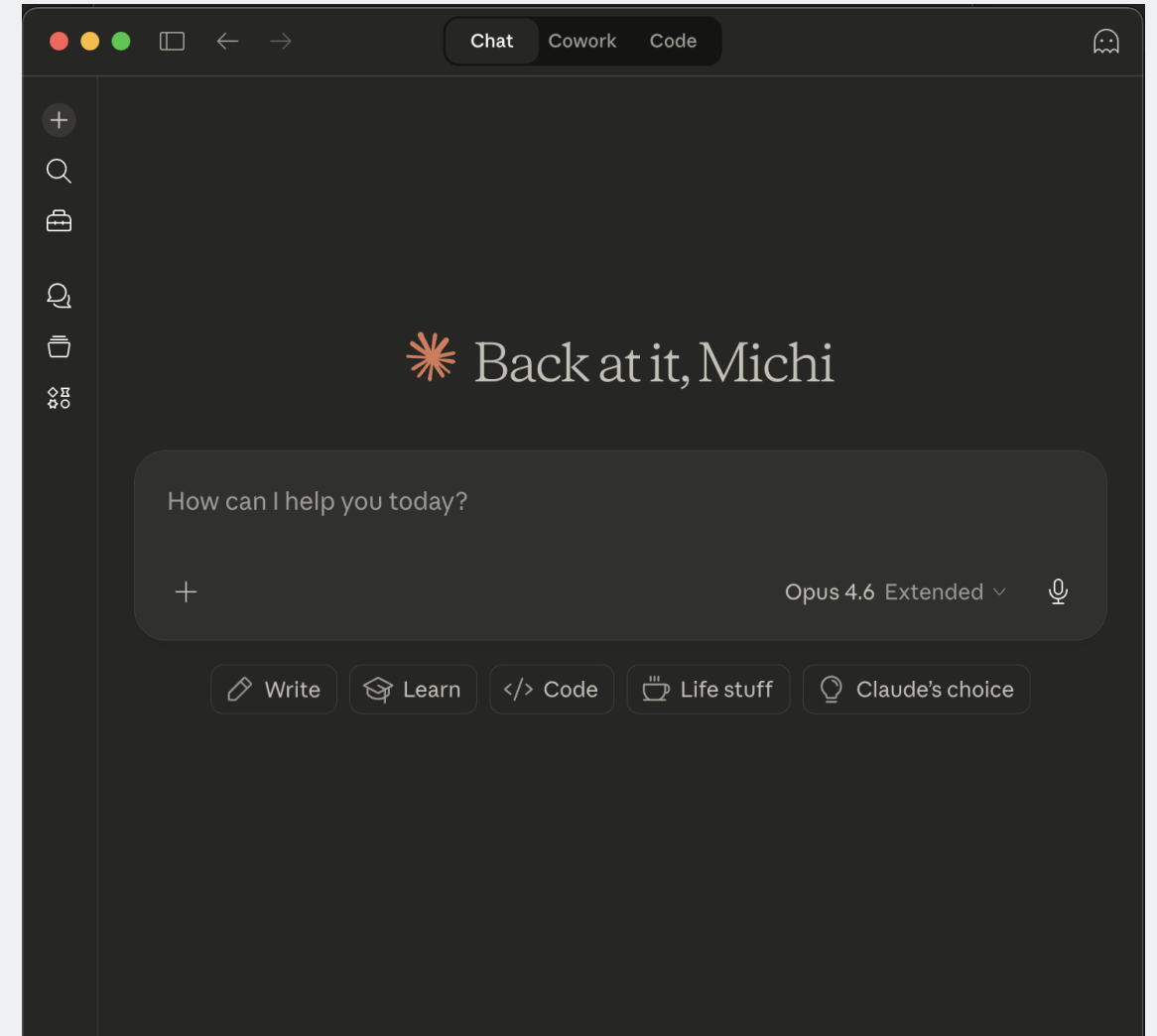
Claude API

Build your own apps

Claude Desktop



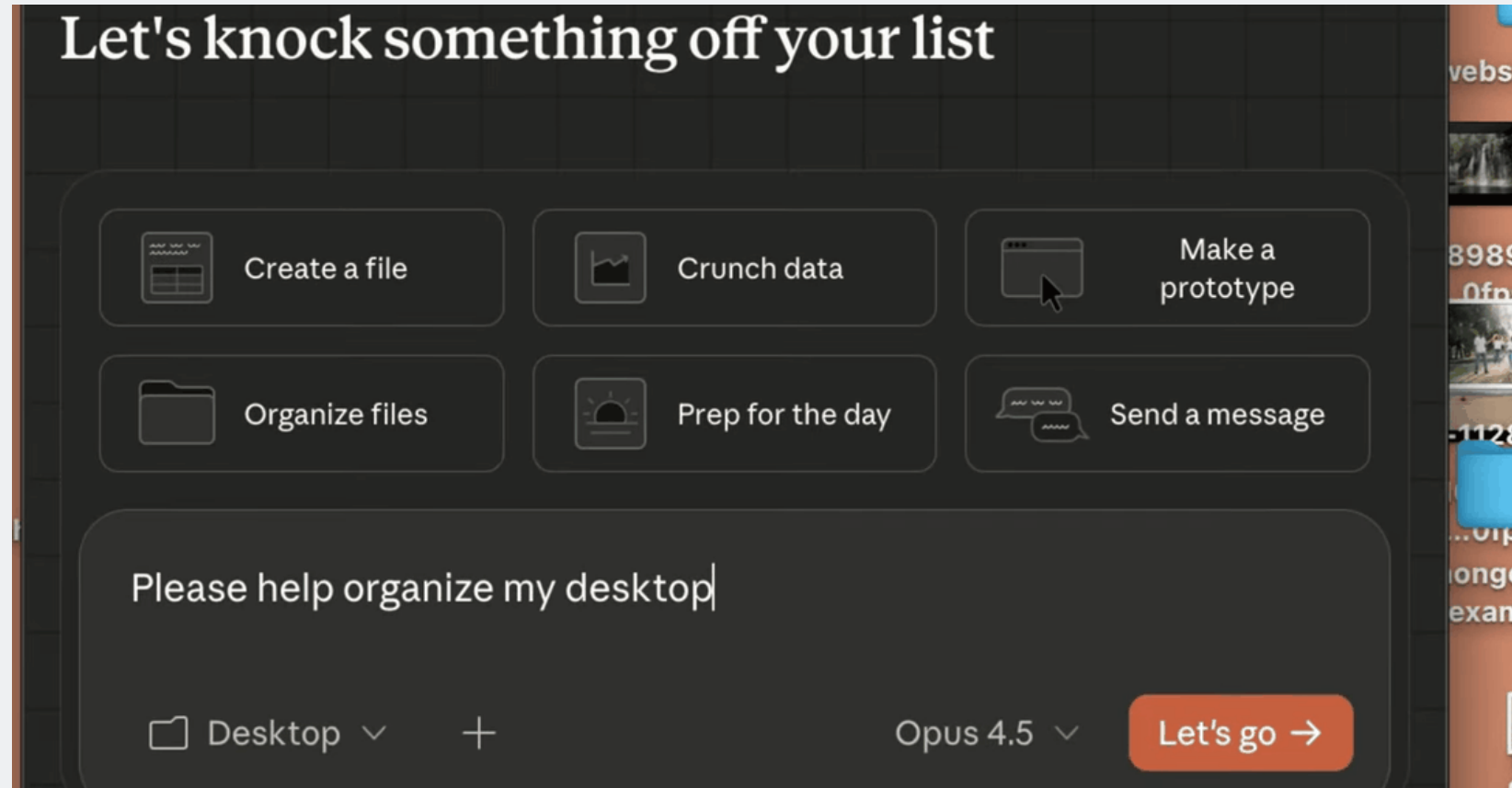
- ▶ Web + mobile chat interface
- ▶ Projects for organized conversations with shared context
- ▶ Artifacts: generates documents, code, diagrams inline
- ▶ Built-in Memory
- ▶ Deep Research for multi-step investigation



Claude Cowork



- ▶ Desktop agent for knowledge workers
- ▶ Controls mouse and keyboard on your computer
- ▶ Automates repetitive desktop workflows
- ▶ Available on Mac + Windows



Claude Code



- ▶ CLI-based AI coding agent in your terminal
- ▶ Reads, writes, and runs code autonomously
- ▶ Skills and Plugins ecosystem
- ▶ "90% of Claude Code's code is written by Claude Code"

The screenshot shows a terminal window titled "Maximize effort and performance" with two tabs: "Investigate system setup and research Claude..." and "Maximize effort and performance (node)". The interface features an orange robot icon and the following text:

```
Claude Code v2.1.92
Opus 4.6 (1M context) · Claude Max
~/Desktop/DS Projects/knowledge_setup
```

The user enters the command `effort /max`, and the AI responds with a bullet point: "Understood – maximum effort mode. What would you like to work on?".

The user enters `hello`, and the AI responds with another bullet point: "Hey Michi! What can I help you with today?".

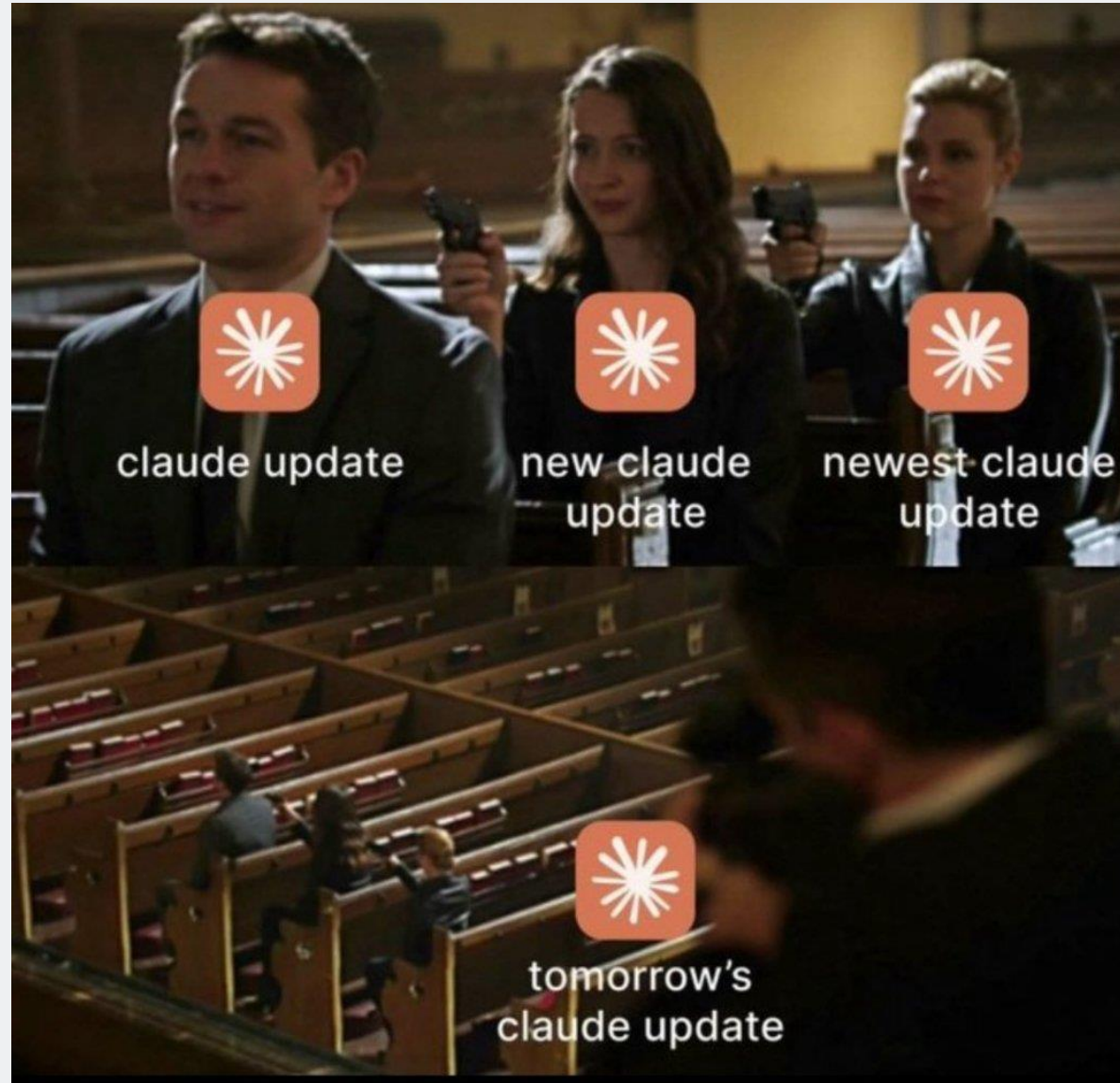
The terminal shows a prompt `>` with a cursor. At the bottom, a status bar displays: `Opus 4.6 | knowledge_setup | main ●2 ↑4 | —...`

Claude API



- ▶ Build AI into your own products
- ▶ Batch API for cheaper offline processing
- ▶ Up to 300K output tokens per request
- ▶ For Business:
 - ▶ Europe Hosted
 - ▶ Zero Data Retention







The Context Window



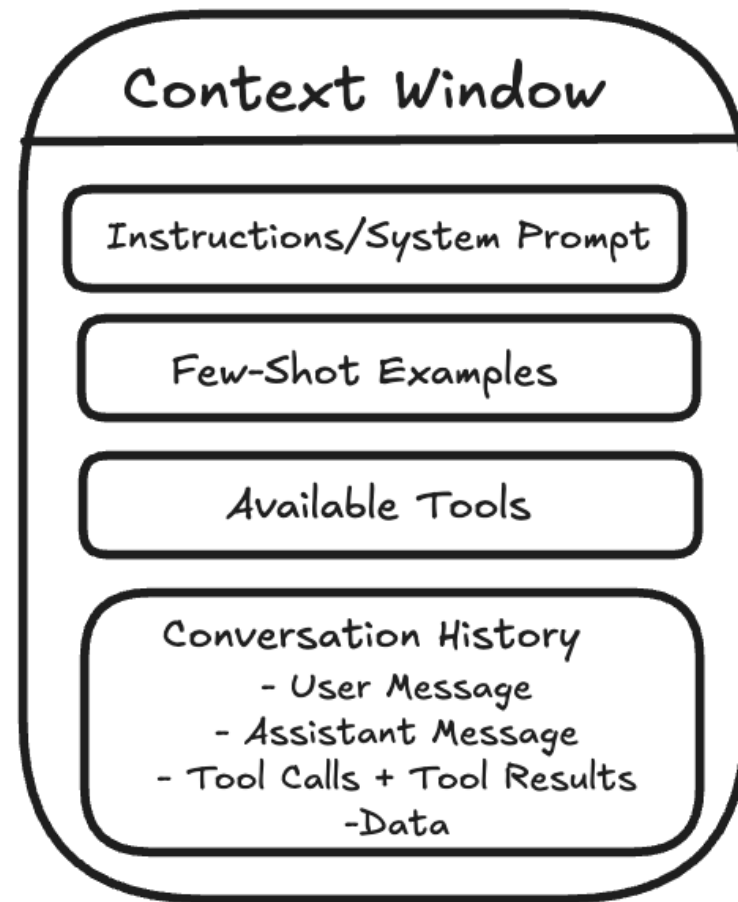
Context Window Basics

Why do LLMs "forget"?

- ▶ No permanent memory by default
- ▶ Each conversation starts fresh
- ▶ Long conversations push early context out

Best practices:

- ▶ Be specific & less is more
- ▶ One task at a time.
- ▶ Clear context or new chat for other tasks





Markdown Files

- ▶ Markdown (.md) = the universal language LLMs understand best
- ▶ Structured text with headers, bullets, tables
- ▶ LLMs parse structured text the way humans scan documents
- ▶ Think of it as "the meta layer" for communicating with AI

```
# 🚀 Markdown: because HTML was  
*too mainstream*
```

```
## 👁️ What you're looking at
```

```
This is Markdown – the "I want formatting but not  
pain" language.
```

```
---  
  
## ✨ Features (aka flexes)
```

- **Bold moves** with `**bold**`
- *Italic vibes* with `*italic*`
- ~~Mistakes~~ with `~~strikethrough~~`
- `code()` for instant +10 dev credibility

CLAUDE.md

Persistent instructions

- ▶ ~/.claude/CLAUDE.md = global (all projects)
- ▶ project/CLAUDE.md = project-level
- ▶ project/CLAUDE.local.md = personal overrides
- ▶ Subdirectory CLAUDE.md = loaded on demand



```
# Project: [NAME]

[One sentence: what this does and who it's for.]

## Tech Stack
- Language: [blank]
- Framework: [blank]
- Database: [blank]
- Frontend: [blank]

## Commands
- Dev server: [blank]
- Run tests: [blank]
- Lint: [blank]
- Build: [blank]
- Type check: [blank or N/A]

## Architecture
[Will be defined during /plan]

## Project Structure
[Will be filled by /setup or /plan]

## Documentation Map
When working on these areas, read the relevant doc BEFORE making changes:
- Architecture: @docs/ARCHITECTURE.md
- Design system: @docs/DESIGN.md
- API contracts: @docs/API.md
- Current plan: @docs/PLAN.md

## Environment
[Will be configured during /setup or /implement]
- Keep .env.example updated when adding new environment variables

## Domain Context
[Will be defined during /plan]

## Known Gotchas
[Add a line every time Claude makes a project-specific mistake]
```




Other Memory Types

RAG (Retrieval Augmented Generation)

- ▶ Store docs, find relevant chunks,
- ▶ AI answers grounded in data

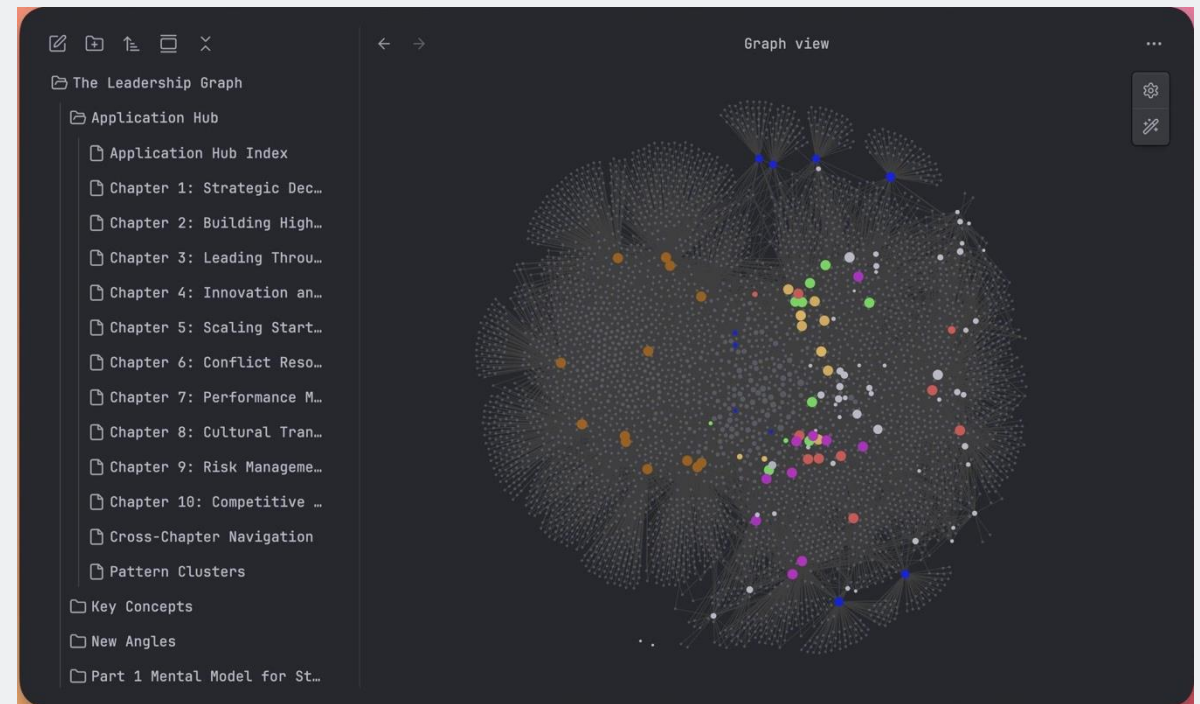
Knowledge Graphs

- ▶ Entities + relationships
- ▶ via MCP

Filesystem

- ▶ Direct file read / write
- ▶ Code Projects, Claude Desktop

Obsidian + Markdown





"Dad why is my sisters name Rose?"

"Because your Mother loves roses"

"Thanks Dad"

"No Problem **CLAUDE**
-DANGEROUSLY-SKIP-PERMISSIONS"



ngflip.com



Skills, MCP and Agents

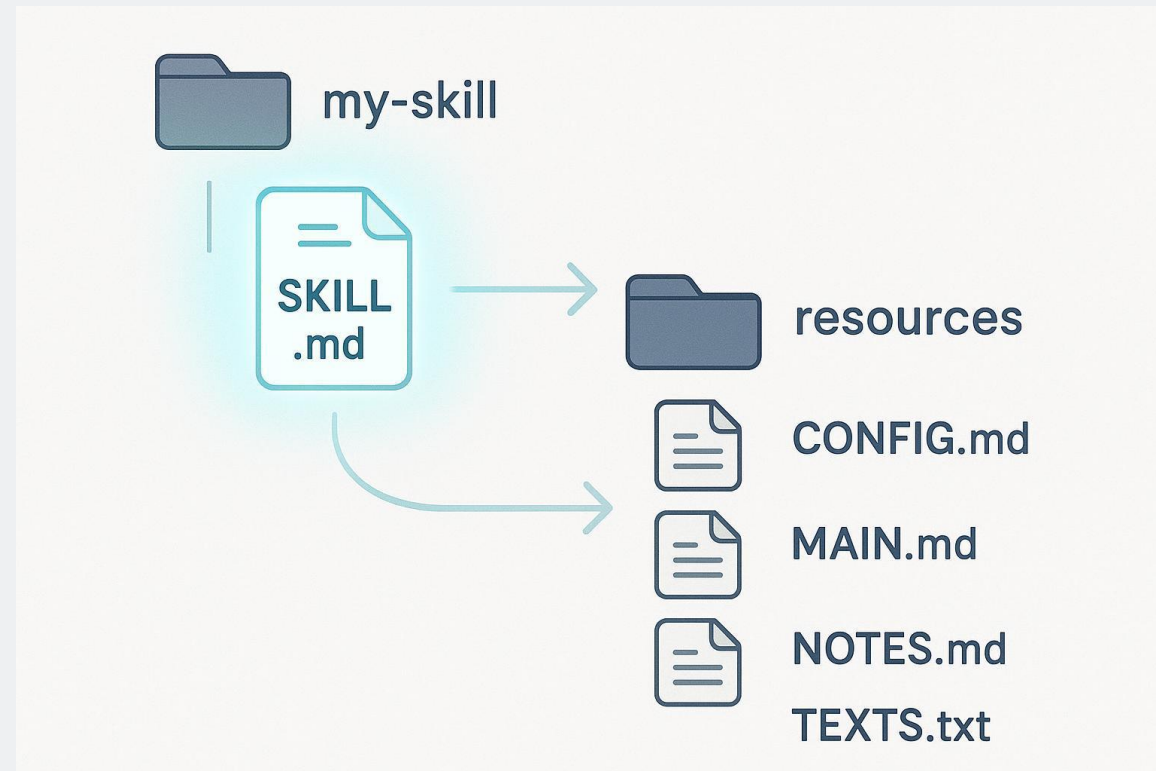


Skills – Reusable Functionalities

- ▶ Markdown files with reusable procedures.
- ▶ Stored in `.claude/skills/`. Claude reads descriptions
- ▶ Loads the full skill when a task matches.

Use cases:

- ▶ Code review workflows
- ▶ Web research delegation
- ▶ Automated documentation generation



MCP's – API for LLM's

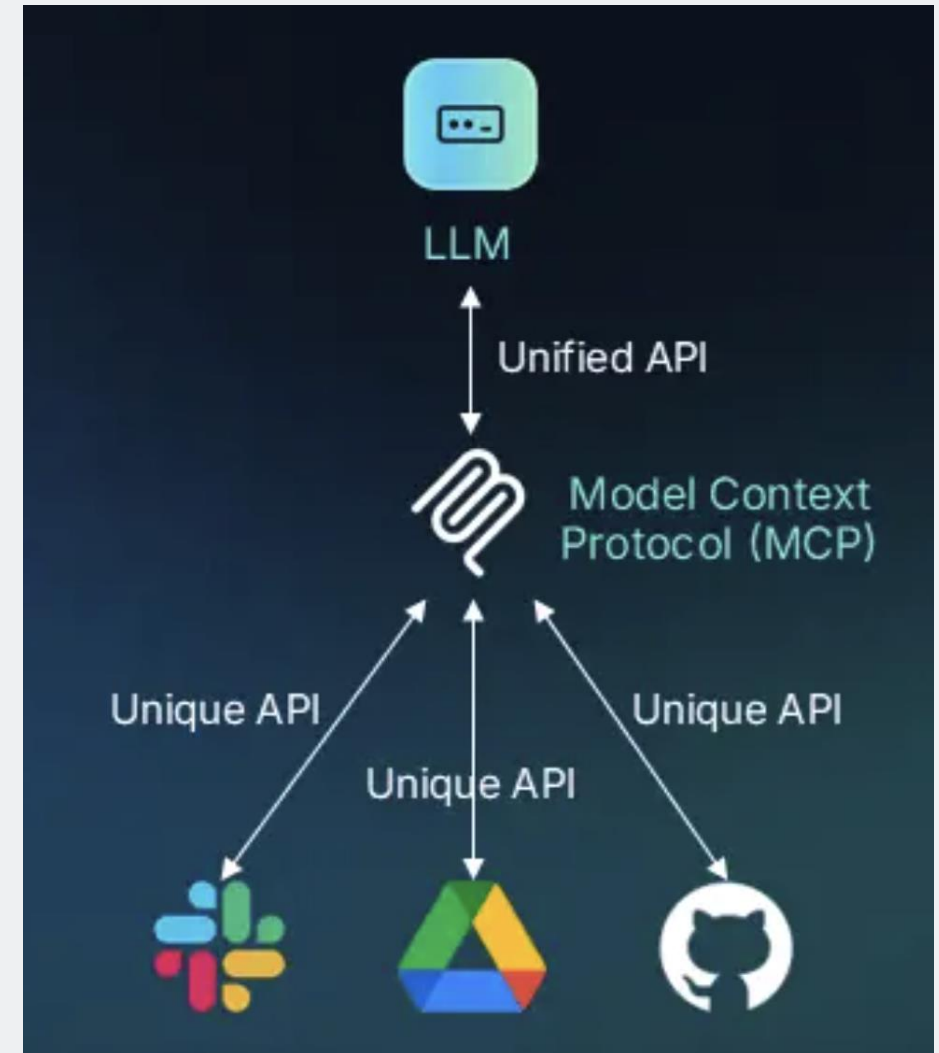


Model Context Protocol

- ▶ Think of it as USB-C for AI: one standard plug for everything.
- ▶ 97 million installs (March 2026)

What you can connect:

- ▶ GitHub, Brave Search, Databases
- ▶ File system, Slack, Google Drive
- ▶ Memory (persistent knowledge graph)



Intro into Agents



What is an agent?

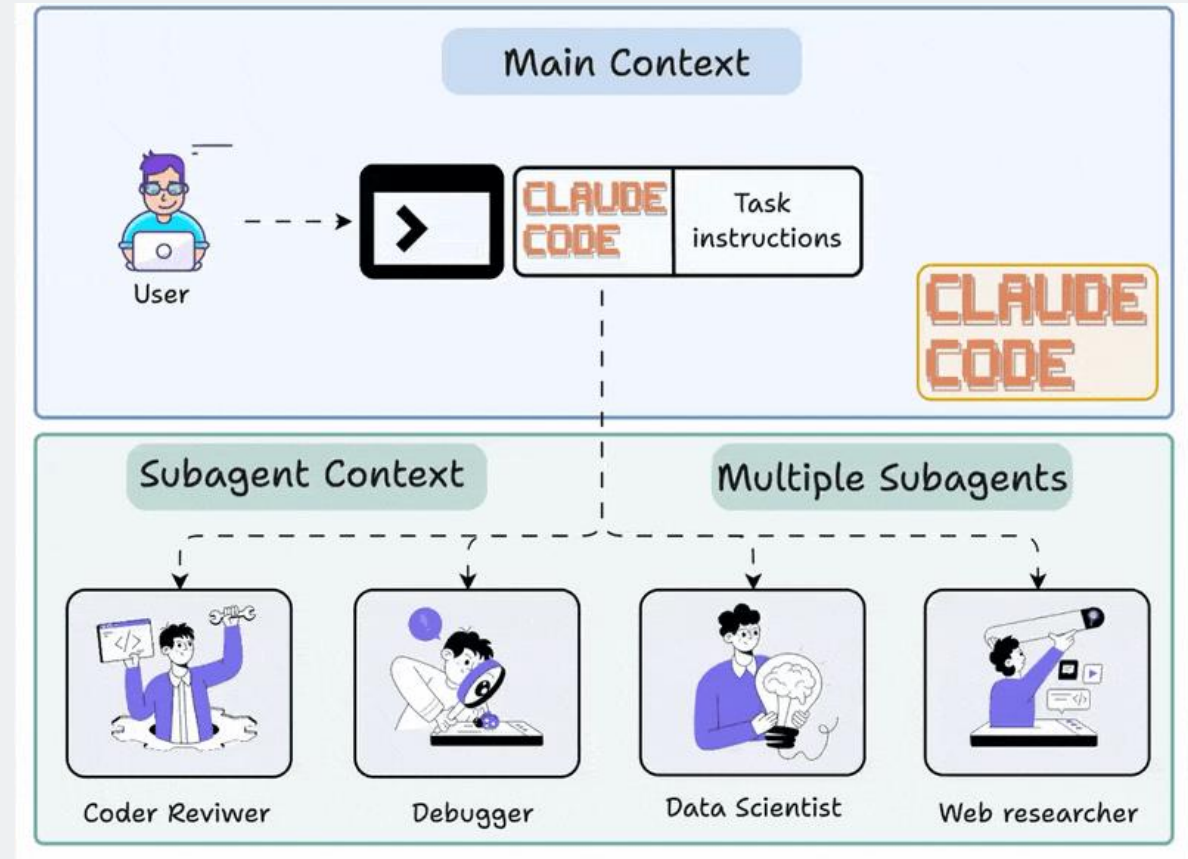
- ▶ An LLM with a system prompt and tools.
- ▶ Plan, act, iterate.

Subagents in Claude Code:

- ▶ Isolated Claude sessions
- ▶ Own context window

Agent Teams:

- ▶ Multiple Agents working in parallel
- ▶ Self Orchestrated





Agents Use Case

Web Researcher Agent

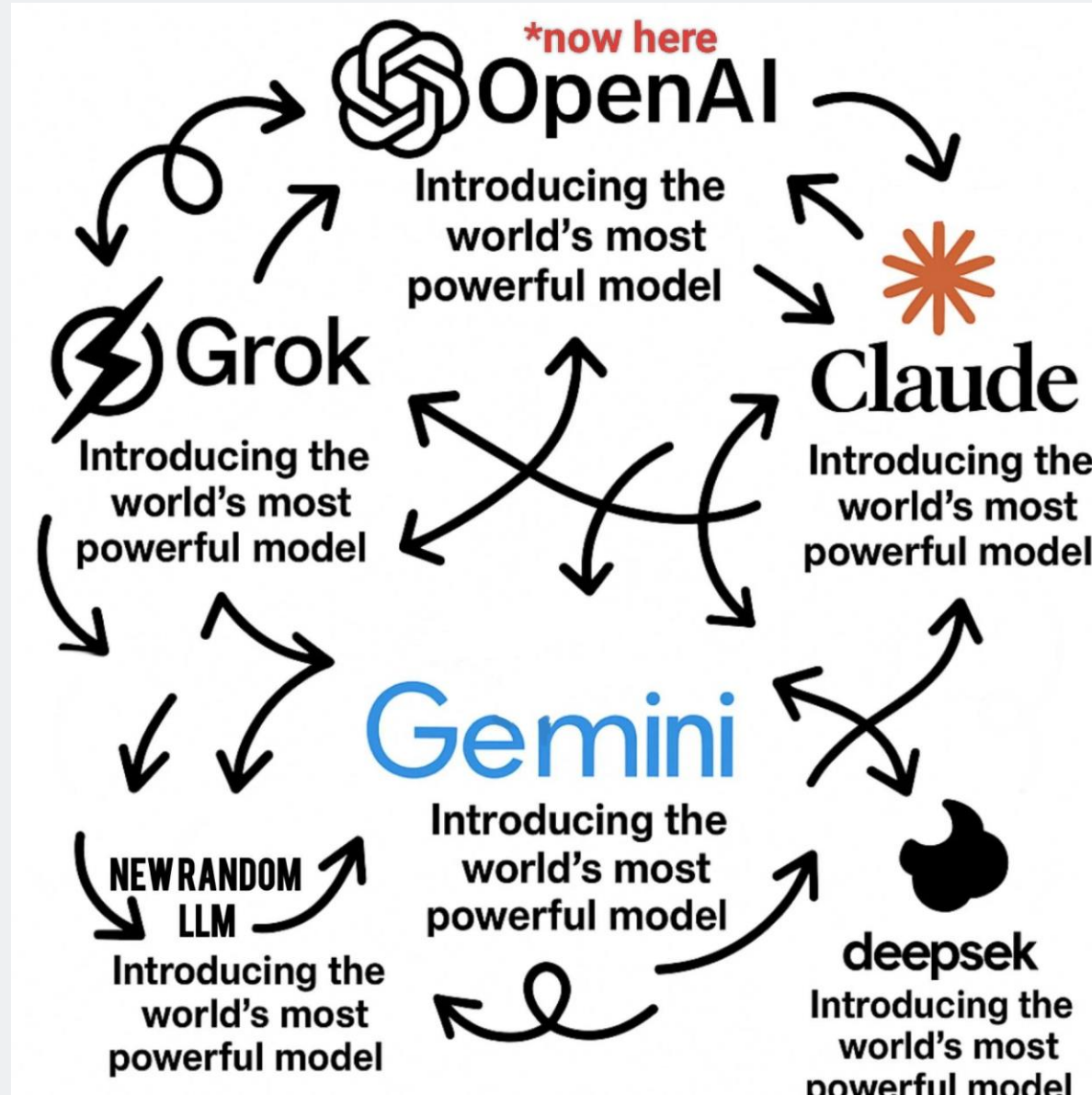
- ▶ Searches the web, follows links, reads pages
- ▶ Returns summarized findings
- ▶ Use: market research, docs lookup, competitive analysis

```
> can you research the web for me with the @"web-researcher
(agent)" for Lumos in Vienna

• web-researcher(Research Lumos in Vienna)
  └─ brave-search - Brave Web Search (MCP)(query: "Lumos
                                           Vienna Austria",
                                           count: 10)

    Running...
    brave-search - Brave Local Search (MCP)(query: "Lumos
                                           Vienna Austria",
                                           count: 10)

    Running...
```



Claude Setup



How to setup Claude Code

- ▶ Give non destructive permissions at user level
- ▶ In the folder you are working give sane editing permissions
- ▶ Require user permission per destructive command in the repo

Workshop



► Manual setup:

- Clone repo: <https://github.com/DominikKanjuh/sane-claude-setup>
- Installing as plugin
 - /plugin marketplace add DominikKanjuh/sane-claude-setup
 - /plugin install sane-claude-setup@sane-claude-setup



Further Resources



Learn on your own (Official)

- ▶ code.claude.com/docs - official docs: best practices, context management, how Claude Code works under the hood
- ▶ anthropic.com/engineering - deep dives from the Claude Code team
- ▶ anthropic.com/news - model releases, changelog, new features
- ▶ Boris Cherny's CLAUDE.md (~100 lines, creator of Claude Code) - the canonical "less is more" example
- ▶ docs.anthropic.com - Claude API reference, for when you want to build your own
- ▶ Anthropic releases RSS feed - subscribe to stay current, the space moves fast



Learn on your own (Community)

- ▶ Shrivu Shankar - blog.sshh.io/p/how-i-use-every-claude-code-feature
- ▶ github.com/hesreallyhim/awesome-claude-code - curated list of skills, hooks, plugins, subagents
- ▶ HumanLayer blog - Writing a good CLAUDE.md
- ▶ r/ClaudeAI and r/ClaudeCode on Reddit - real workflows, real gotchas
- ▶ Anthropic Discord - discord.gg/anthropic - fastest way to get unstuck
- ▶ Follow @bmcherny and the Anthropic team on X for workflow tips

**ZERO
ONE
HACK
AI HACKATHON**

**MAY 29 - 31
2026**

**VIENNA,
AUSTRIA**



WHAT IS ZERO ONE HACK?

36-hour build sprint in Vienna with real-world challenges

ZERO
ONE
HACK
AI HACKATHON

THEME

AI Sovereignty of EU

Innovative Solutions
for Europe's Industrial
Future

Strengthen
Austrian Tech
Competitiveness



HARDWARE

Participants will get access to Compute provided by HPE and EuroHPC to run & train own models.

HPE

16x NVIDIA H200NVL GPUs
with 2.2TB total VRAM

LEONARDO
CINECA

NVIDIA A100 GPUs
provided by AI:AT

EVENT OVERVIEW

**ZERO
ONE
HACK**
AI HACKATHON

Friday, May 29 - Opening Night

19:30 Opening Ceremony
20:00 Partner Presentations
21:00 Case Reveal
22:00 Hacking Begins

Saturday, May 30 - Hacking Day

11:00 Panel Discussion
13:00 Mentoring & Workshops
19:00 Social Activity
Overnight Hacking

Sunday, May 31 - Closing Ceremony

10:00 Submissions & Speakers
10:30 Team Pitches & Judging
13:30 Winners Announced
14:00 Networking

KEY FACTS

- 80+ selected participants
- 3 specialized tracks
- 6 expert judges
- 5+ mentors
- Professional content creation



AI:AT FACTORY

100+ Capacity
Event Space
WIFI + Power
Food & Drinks

Sign up here

Early Access: Be the first to register to our Hack

ZERO
ONE
HACK
AI HACKATHON



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